

PROBLEM-SOLUTION FIT

Project Title

Rising Waters

Project Description

Rising Waters is an AI-powered Flood Prediction System developed using Machine Learning. The system predicts flood risk using rainfall and environmental parameters, helping authorities issue early warnings and improve disaster preparedness.

Introduction

Floods are one of the most destructive natural disasters, causing significant loss of life, property, and infrastructure every year. Traditional forecasting methods often fail to provide accurate and timely predictions.

Problem Statement

- Delayed flood warnings.
- Inaccurate flood prediction.
- Difficulty in disaster preparedness.
- Loss of lives and property.
- Lack of intelligent prediction systems.

Proposed Solution

The Rising Waters application uses Machine Learning algorithms to analyze rainfall and weather conditions and predict flood risk. The system provides instant results through a user-friendly web application.

Why This Solution?

- Provides accurate flood prediction.
- Supports early warning systems.
- Helps disaster management authorities.
- Reduces response time.
- Improves public safety.

Features

- Machine Learning prediction.
- User-friendly dashboard.
- Real-time flood risk analysis.
- Fast prediction.
- Easy deployment using Streamlit.

Expected Benefits

- Better disaster management.
- Early evacuation planning.

- Reduced flood damage.
- Improved decision making.
- Reliable prediction results.

Conclusion

The Rising Waters system effectively addresses the limitations of traditional flood prediction methods by using Artificial Intelligence and Machine Learning to provide faster and more accurate flood risk predictions.